

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 30

INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 11

CRYPTOCURRENCY

- A cryptocurrency is a virtual currency that is secured by cryptography, which makes it nearly impossible to tamper with.
 - ✓ Virtual currency is a type of unregulated digital currency that is only available in electronic form. It is stored and transacted only through designated software, mobile or computer applications, or through dedicated digital wallets, and the transactions occur over the internet through secure, dedicated networks.
 - ✓ Virtual currency or virtual money, is a type of unregulated digital currency, which is issued and controlled by its developers and used and accepted among the members of a specific virtual community.
 - ✓ Virtual currency is a subset of digital currency.



CRYPTOCURRENCY

- Cryptocurrencies are decentralized networks based on Blockchain technology.
- A defining feature of cryptocurrencies is that they are not issued by any central authority, making them immune to government interference or control.
- Cryptocurrencies are systems that allow for secure payments online which are expressed in terms of virtual 'tokens' which are represented by ledger entries to the system.
- Cryptocurrencies make it easier to transfer funds directly between two parties, without the need for a trusted third party like a bank or credit card company.
- 'Crypto' refers to the various encryption algorithms and techniques that safeguard these entries (example - hashing). Some of the cryptography used in cryptocurrency today was originally developed for military applications.
- The first blockchain-based cryptocurrency was Bitcoin, which still remains the most popular and most valuable.
- For transacting on the Bitcoin network, the users must run a program called 'Wallet'. Each wallet has two unique and distinct cryptographic keys – a public key and a private key.



INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

CRYPTOCURRENCY

- The public key is generated from the private key using a complex algorithm but not vice versa. The public key appears on the Blockchain ledger as the user's digital signature and Bitcoins can be deposited using the public key, but cannot be withdrawn without the private key. If any user misplaces his private key, he will not be able to access his Bitcoin wallet.
- Today, there are thousands of alternate cryptocurrencies with various functions and specifications.
- Bitcoin was launched in 2009 by an individual or group known by the pseudonym "Satoshi Nakamoto."

Bitcoin Genesis Block

Raw Hex Version

```

00000000 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000020 00 00 00 00 3b a3 bd fd 7a 7b 12 b2 7a c7 2c 3e ....)tjz{.*q.>
00000030 47 74 8f 41 7f c8 19 c3 88 8a 51 32 3a 9f 98 aa qv.a.B.A"5Q2rY.*
00000040 48 1e 5e 4a 29 a9 5f 49 ff ff 00 10 10 ac 28 7c K."2}.*_py...~*|
00000050 01 01 00 00 00 02 00 00 00 00 00 00 00 00 00 00 .....
00000060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000070 00 00 00 00 00 00 ff ff ff ff 4d 04 ff ff 00 10 .....ffff~ff..
00000080 01 04 45 54 68 65 20 54 69 4d 45 73 20 30 30 2f ..EThe Times 03/
00000090 4a 41 4e 2f 32 30 30 39 20 43 48 61 6e 63 65 6c Jan/2009 Chancel
000000a0 4c 4f 72 20 4f 4e 20 42 72 49 4e 4b 20 4f 44 20 lor on bris of
000000b0 73 45 43 4f 6e 64 20 42 61 49 4c 4f 73 74 20 46 second bailout f
000000c0 6f 72 20 42 61 6e 63 73 ff ff ff ff 01 00 f2 05 or bankruptcy...b.
000000d0 2a 01 00 00 00 43 41 04 67 8a fd 30 fe 55 48 27 *....CA.s5g'you'
000000e0 19 47 f1 a6 71 30 37 10 5c 04 a8 28 80 39 09 a6 .qhiq!+.!0'(kp.
000000f0 79 42 20 8a 1f 61 de 86 49 f6 9c 3f 4c 3f 38 04 y0M.abY10A711RA
00000100 f3 55 04 85 1e c1 12 08 5c 38 4d ft 8a 08 8d 57 0U.&.A.9\8n+9..W
00000110 8a 4c 70 2b 4b f1 10 5f ac 00 00 00 00 00 00 00 00 5cp+k8_~....

```

Satoshi Nakamoto message
embedded in the coinbase of the
first block

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

CRYPTOCURRENCY

- As of March 2021, there were over 18.6 million Bitcoin in circulation with a total market value of around \$927 billion.
- Some of the competing cryptocurrencies spawned by Bitcoin's success, known as 'altcoins' include Litecoin, Peercoin, Dogecoin, Ethereum, Cardano and EOS.
- Today, the aggregate value of all the cryptocurrencies in existence is around \$1.5 trillion and Bitcoin currently represents more than 60% of the total value.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299